

Test 2 - High-Price Discharge Test

Purpose

The purpose of this test is to verify that the EMS follows the correct **high-price discharge logic** when real load data and real API price data are used.

The test should confirm that:

Condition	Expected scenario
HIGH price + PV available	S4
HIGH price + no PV + battery usable	S5
HIGH price + no PV + battery unusable + PEM SoC > 10% + load < 40 mW	S6
Otherwise	S1

The main transitions to observe are:

S4 → **S5**, when PV becomes unavailable

S5 → **S6**, when the virtual Battery SoC reaches the lower threshold and Power is lower than 40mW

S6 → **S1**, if PEM becomes unavailable or the load is too high

Initial setup

Using Scripts:

AppLab: ems_base_estimation_logger_v5

Arduino IDE: ControllableLoad_V4.1_triggerAndData

Item	Value
Test duration	24 minutes
Simulated duration	24 hours
Price data	Real API day with HIGH prices
Load profile	Active, 15–75 mW
Initial Battery SoC	20%

Initial PEM SoC	100%
PV/lamp at start	Off at start, then On from 6:00 to 18:30
CSV logging	Enabled

Demo and EMS control

Start the EMS system first. Then start the demo to reset the 96-slot timeline, begin CSV logging and activate the variable load trigger. Raw logging can run independently of the demo.

Reset

Start EMS system

Stop EMS system

Start demo + logging

Stop demo

Auto mode

Manual mode

Start raw log

Raw log

Stopped

Raw log file

-

Raw samples

0

Last raw update

-

Price test date

07/05/2026

Manual electricity price

0.99064

Hydrogen estimation (%)

100.0

Virtual battery charge state (%)

20.0

Load prices for date

Apply manual price

Initiate H2 estimation manually

Apply virtual battery state

Control mode

auto

Demo status

Stopped

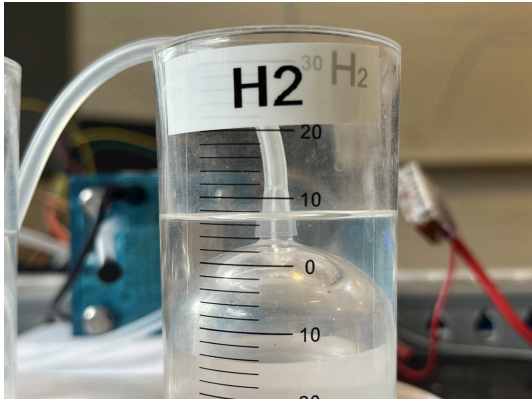
Current slot

0

Scenario

S1

Storage states		INA sensor measurements	
Current battery voltage	3.689 V	PV voltage	2.967 V
Initial SoC lookup voltage	3.695 V	Battery voltage	3.689 V
Lookup-estimated real battery SoC	4.6 %	PEM voltage	1.037 V
Integrated real battery SoC	4.7 %	Load voltage	5.117 V
Integrated real battery charge	93.759 mAh / 2000.0 mAh	PV current	0.03 mA
Virtual battery SoC	20.0 %	Battery current	-0.03 mA
Virtual battery capacity	10.00 mAh	PEM current	-0.04 mA
Virtual battery charge	2.000 mAh	Load current	-0.01 mA
Virtual battery state	low	PV power	0.07 mW
Hydrogen volume	15.60 mL	Battery power	-0.09 mW
Usable hydrogen	8.00 mL	PEM power	-0.04 mW
Hydrogen estimation	100.0 %	Load power	-0.04 mW
Hydrogen mode	idle	Load trigger	LOW
		Arduino mode	S1 Grid -> Load



Hydrogen Niveau ved start af test (100% = 8mL) (efter opladning med system)

Test procedure

1. Load the selected real API price day with HIGH prices.
2. Set the initial Battery virtual SoC.
3. Set the initial PEM virtual SoC.
4. Start the load profile.
5. Turn the lamp on before starting the test.
6. Start CSV logging.
7. Start the EMS test.
8. Light starts OFF until 06:00 (**S5** is should be selected while PV off)
9. Confirm that the EMS selects **S4** while PV is available.
10. Turn the lamp off after the planned PV period.
11. Observe whether the EMS switches to **S5** when PV becomes unavailable.
12. Continue the test until the Battery virtual SoC reaches the lower threshold.
13. Observe whether the EMS switches to **S6** when Battery is unusable, PEM SoC is above 10%, and load is below 40 mW.
14. Observe whether the EMS switches to **S1** when PEM cannot be used.
15. Stop the test after 24 minutes.

CSV file

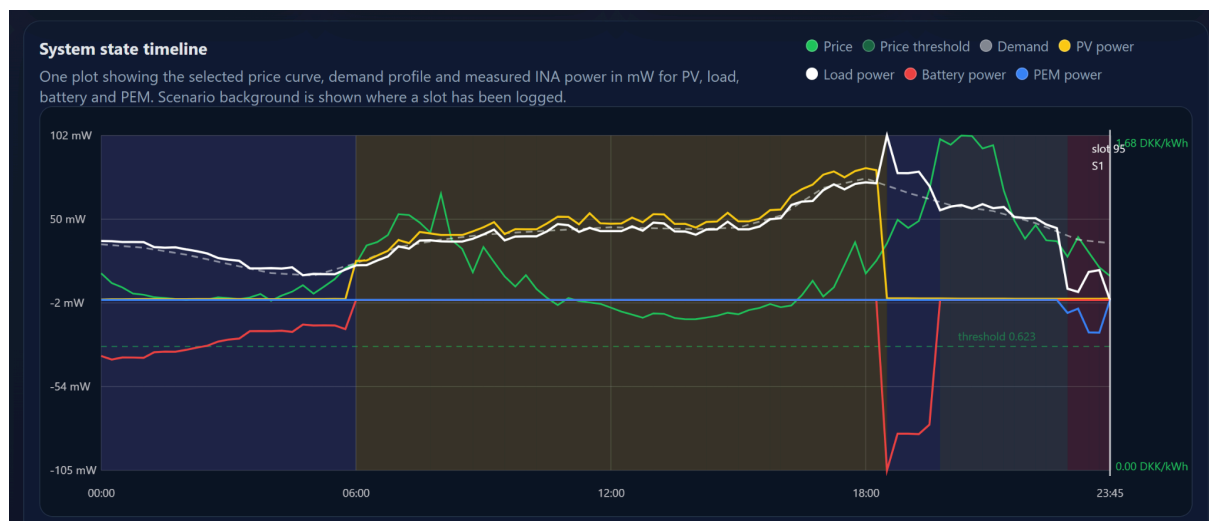
Item	Entry
CSV filename	/app/python/logs/ems_demo_20260512_133302.csv
Saved location	

Notes

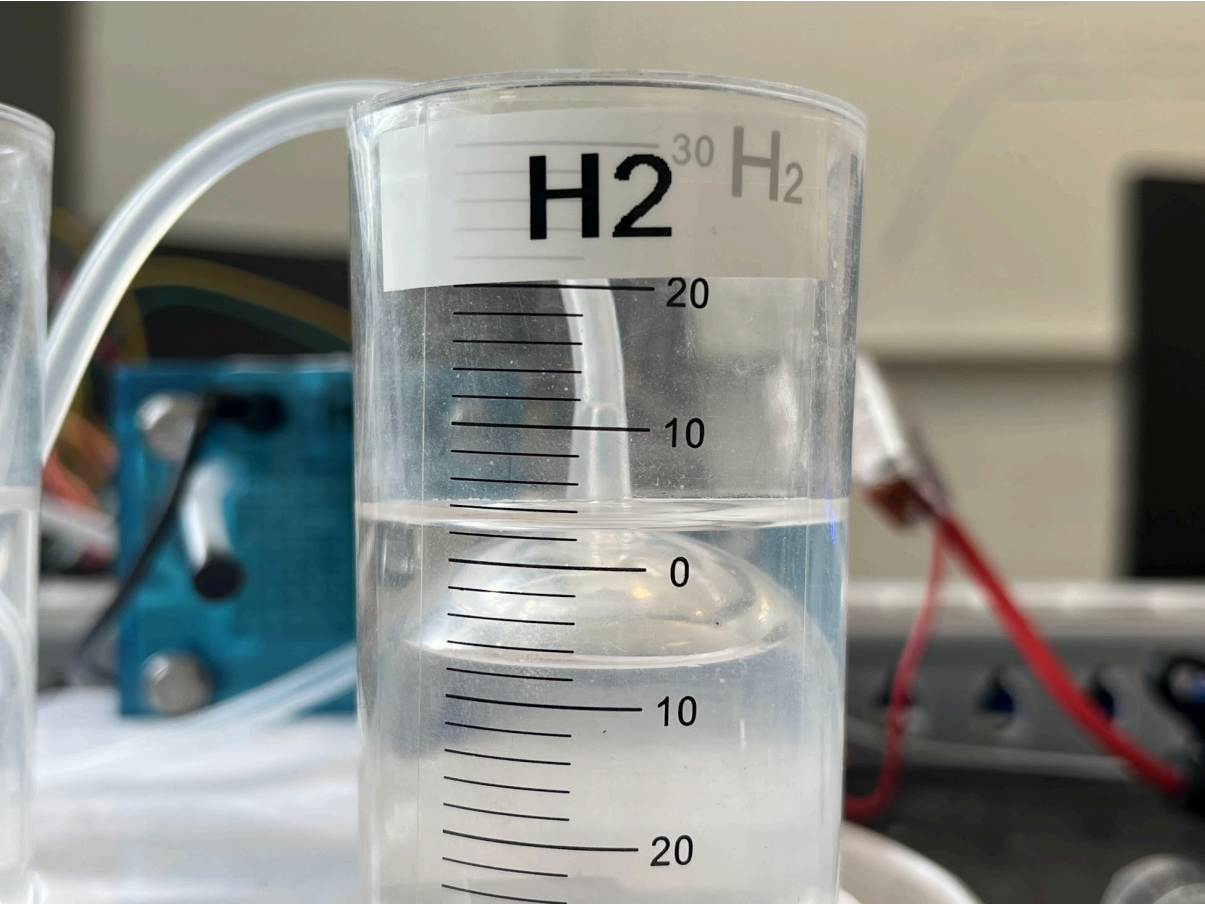
Time / timestep	Observation
00:00	S5 Batt to load, no PV available
06:00	Lamp is switched off
06:00 - 18:30	S4 PV to Load, since PV became available
18:30	PV is switched off
22:45	S6 PEM to Load, since the Demand is again below 40mW

Result summary

Question	Answer
Did the system select S4 when PV was available?	Yes at 06:00 the system switched to S4
Did the system switch to S5 when PV became unavailable?	This state was chosen initially (00:00 to 06:00)
Did Battery SoC decrease during S5?	Yes the Battery SoC decreased gradually during S5
Did the EMS switch to S6 when Battery SoC reached the lower threshold?	The PEM was never switched on as the Demand was too high
Did PEM only operate when load was below 40 mW?	The PEM was never switched on as the Demand was too high
Did the system fall back to S1 when PEM could not be used?	System went to S1 when battery was discharged and load was still above 40mW
Any unexpected scenario changes?	
Overall result	Pass / Partial / Fail



Storage states		INA sensor measurements	
Current battery voltage	3.693 V	PV voltage	3.110 V
Initial SoC lookup voltage	3.695 V	Battery voltage	3.693 V
Lookup-estimated real battery SoC	4.7 %	PEM voltage	0.857 V
Integrated real battery SoC	4.6 %	Load voltage	5.125 V
Integrated real battery charge	92.614 mAh / 2000.0 mAh	PV current	0.15 mA
Virtual battery SoC	10.0 %	Battery current	-0.03 mA
Virtual battery capacity	10.00 mAh	PEM current	-0.04 mA
Virtual battery charge	0.997 mAh	Load current	-0.01 mA
Virtual battery state	empty	PV power	0.47 mW
Hydrogen volume	15.18 mL	Battery power	-0.09 mW
Usable hydrogen	7.58 mL	PEM power	-0.04 mW
Hydrogen estimation	94.7 %	Load power	-0.04 mW
Hydrogen mode	idle	Load trigger	LOW
		Arduino mode	S1 Grid -> Load



Hydrogen Niveau after test (95% = 7.58mL)